MECH321
Dynamics of Engineering Systems

Lecture 2 Laplace Transform

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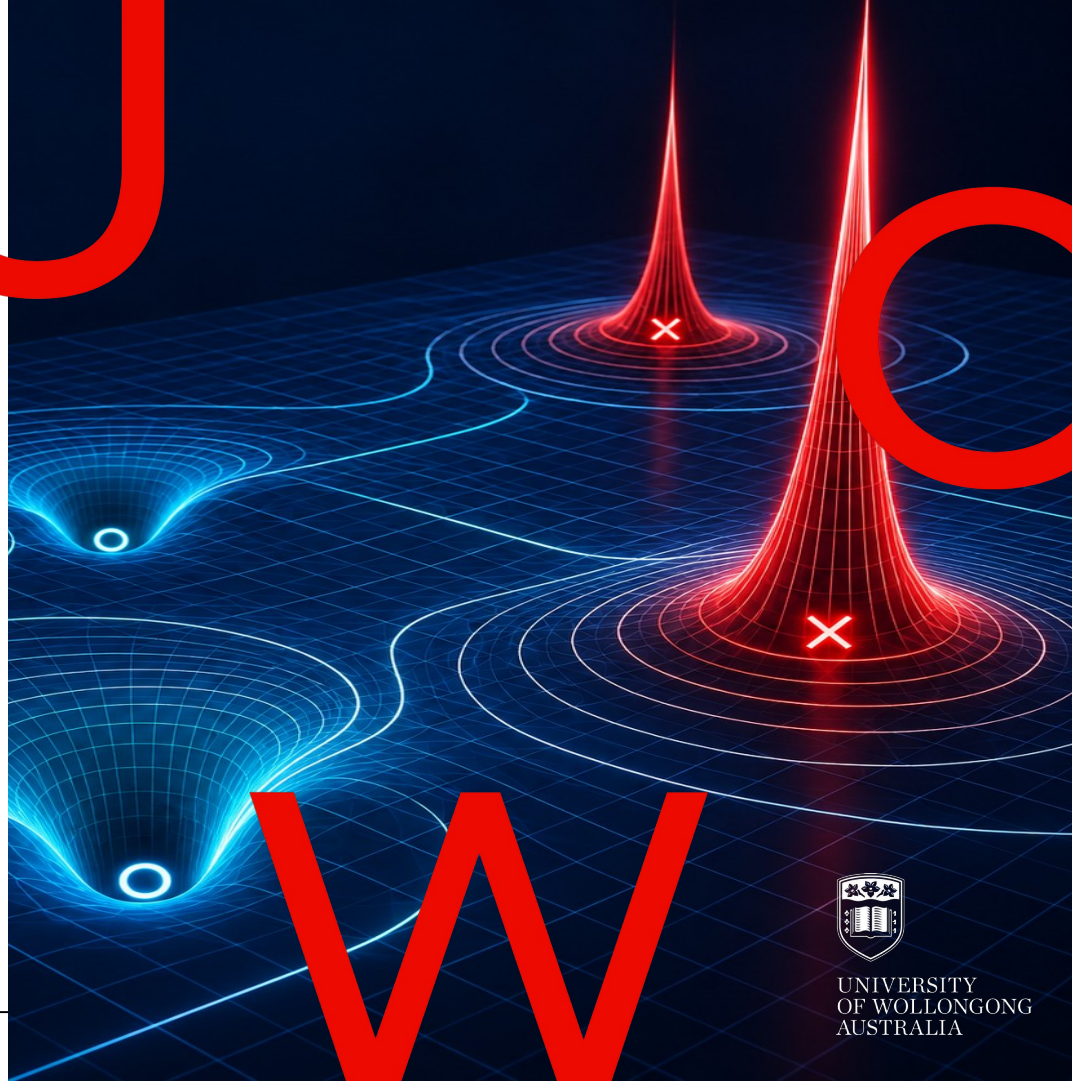


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Initial conditions in the s-domain

PART 1

Forward Laplace transform: ODE to algebra

01 Why transform an ODE?

02 Transform pairs, both shifts, pulse and impulse

03 Derivatives and integration

Why use the Laplace transform?

ODE → algebraic equation (LTI systems)

TIME DOMAIN

$$A\frac{dh}{dt} + \frac{h}{R} = q_{\text{in}}(t)$$

Unknown $h(t)$ and its derivative; initial level $h(0)$ is given.

Laplace transform

s-DOMAIN

$$\left(As + \frac{1}{R}\right)H(s) = Q_{\text{in}}(s) + Ah(0)$$

Solve an algebraic equation for $H(s)$; $h(0)$ remains visible.

1 MODEL

ODE + initial state

2 TRANSFORM

Equation in $H(s)$

3 SOLVE

Find $H(s)$

4 INVERT

Recover $h(t)$

Check $h(t)$ against the ODE and initial condition.

Laplace transform: definition

ONE-SIDED TRANSFORM

$$\mathscr{L}\{f(t)\} = F(s) = \int_{0^-}^{\infty} f(t)e^{-st}\,dt$$

$s = \sigma + j\omega$ (σ: exponential weighting; ω: oscillation rate)

For each s, $F(s)$ is the weighted integral of $f(t)$ over $t \geq 0$; varying s gives the transform function.

Derivative: $sF(s) - f(0^-)$. Requires convergence at the chosen s.

Laplace transform: linearity

THE PROPERTY

$$\mathscr{L}\{af(t)+bg(t)\}=aF(s)+bG(s)$$

Example:

$$m\ddot{x}+b\dot{x}+kx=F_1(t)+F_2(t)$$

$$X(s)=\frac{F_1(s)+F_2(s)}{ms^2+bs+k}=\underbrace{\frac{F_1(s)}{ms^2+bs+k}}_{X_1(s)}+\underbrace{\frac{F_2(s)}{ms^2+bs+k}}_{X_2(s)}$$

Solve the step and impact responses separately, then add their displacements.

Add zero-state responses. Include the initial-state response once.

Laplace transform: standard pairs

$f(t),\ t \geq 0$	$F(s)$
$\delta(t)$	1
1	$\frac{1}{s}$
t	$\frac{1}{s^2}$
e^{at}	$\frac{1}{s-a}$
te^{at}	$\frac{1}{(s-a)^2}$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{-at}\sin(\omega t)$	$\frac{\omega}{(s+a)^2+\omega^2}$

$$\mathscr{L}\{1\} = \int_0^\infty e^{-st}\,dt = \left[-\frac{e^{-st}}{s}\right]_0^\infty = \frac{1}{s}$$

$$\mathscr{L}\{e^{at}\} = \int_0^\infty e^{-st}e^{at}\,dt = \int_0^\infty e^{-(s-a)t}\,dt \qquad = \left[-\frac{e^{-(s-a)t}}{s-a}\right]_0^\infty = \frac{1}{s-a}$$

$$\mathscr{L}\{te^{at}\} = \int_0^\infty te^{-(s-a)t}\,dt, \quad b = s-a$$

$$u = t, \quad dv = e^{-bt}dt, \quad v = -\frac{e^{-bt}}{b}$$

$$= \left[-\frac{te^{-bt}}{b}\right]_0^\infty + \frac{1}{b}\int_0^\infty e^{-bt}\,dt \qquad = 0 + \frac{1}{b}\cdot\frac{1}{b} = \frac{1}{(s-a)^2}$$

Exponential multiplication: the s-shift

CORE

THE PROPERTY: MULTIPLY THE TIME SIGNAL BY e^{-at}

$$F(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$$

$$\mathscr{L}\left\{e^{-at}f(t)\right\} = \int_0^\infty f(t)e^{-(s+a)t}\,\mathrm{d}t = F(s+a)$$

Decay e^{-at} and weighting e^{-st}: replace s by $s + \mathrm{a}$ in the transform of $f(t)$.

WORKED EXAMPLE: DAMPED COSINE

$$\mathscr{L}\{\cos(2t)\} = \frac{s}{s^2+4} \longrightarrow \mathscr{L}\left\{e^{-2t}\cos(2t)\right\} = \frac{s+2}{(s+2)^2+4}$$

Delayed input

A time shift introduces an exponential factor in the s-domain

THE SAME INPUT, STARTED LATER

Top: start at $t = 0$. Bottom: start at $t = t_d$.

$$F_0 u(t - t_d) \quad \longleftrightarrow \quad \frac{F_0 e^{-st_d}}{s}$$

NUMERICAL EXAMPLE

If $F_0 = 2$ and $t_d = 1.5$, the input transform is $\frac{2e^{-1.5s}}{s}$. Compute the ordinary step response first; then shift its graph right by 1.5 seconds.

$$g(t - t_d)u(t - t_d) \quad \longleftrightarrow \quad e^{-st_d} G(s)$$

Shift the whole function: $g(t)u(t - t_d)$ is usually a different signal.

Force begins at $t = 0$



Force begins at $t = t_d$



Unit impulse

A unit-area pulse approaches an ideal instantaneous input

UNIT-AREA PULSE

The pulse lasts Δ seconds. Set its height to 1/Δ, so its **area remains 1** as Δ shrinks.

$$p_\Delta(t) = \frac{1}{\Delta}[u(t) - u(t-\Delta)]$$

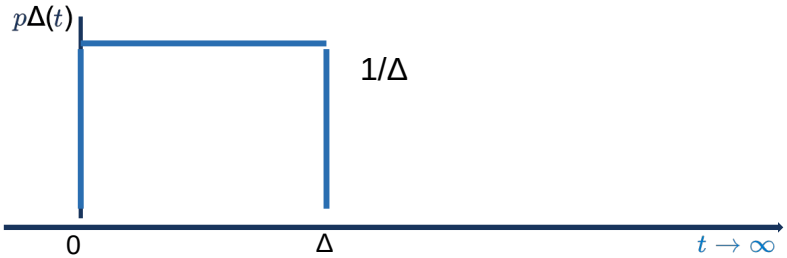
$$P_\Delta(s) = \frac{1 - e^{-s\Delta}}{s\Delta}$$

UNIT IMPULSE LIMIT

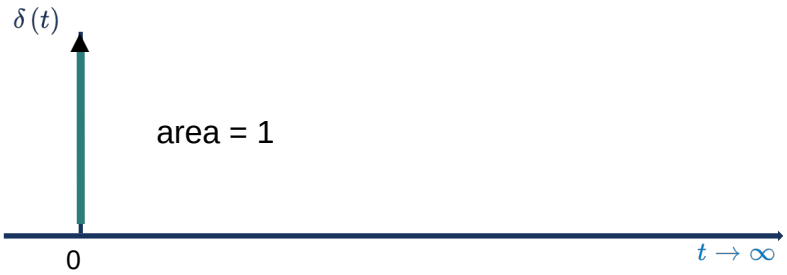
$$P_\Delta(s) \to 1 = \mathscr{L}\{\delta(t)\}$$

Impulse $\delta(t)$: area 1; transform 1.

Rectangular unit-area pulse



Unit impulse as Δ → 0



Worked transform: both shift properties

CORE

Combine a decaying cosine with a delayed step

QUESTION: FIND $F(s)$ AND CHECK f(0⁺)

$$f(t) = 1 + 3e^{-2t}\cos(2t) + 2u(t-2), \qquad t \geq 0$$

STEP 1: TRANSFORM EACH TERM

$$\mathscr{L}\{1\} = \frac{1}{s} \qquad \mathscr{L}\{e^{-2t}\cos(2t)\} = \frac{s+2}{(s+2)^2+4} \qquad \mathscr{L}\{2u(t-2)\} = \frac{2e^{-2s}}{s}$$

STEP 2: ADD THE THREE TRANSFORMS

$$F(s) = \frac{1}{s} + \frac{3(s+2)}{(s+2)^2+4} + \frac{2e^{-2s}}{s}$$

CHECK THE START AND THE DELAY

At $t = 0^+$, $f(0^+) = 4$: the delayed step is still off. ($\lim_{s\to\infty} sF(s) = 1+3+0 = 4$), At $t = 2$, the input jumps by 2.

Laplace transform: derivatives

Initial position and velocity enter the transformed mechanical equation

DERIVATIVES CARRY INITIAL CONDITIONS

$$\mathscr{L}\{\dot{x}\} = sX(s) - x_0$$

$$\mathscr{L}\{\ddot{x}\} = s^2X(s) - sx_0 - v_0$$

Here $x_0 = x(0^-)$ and $v_0 = \dot{x}(0^-)$. The boundary terms arise when the transform integral is integrated by parts; they **cannot be discarded unless the initial state is zero**.

APPLY BOTH RULES TO THE MECHANICAL MODEL

$$m\ddot{x} + b\dot{x} + kx = F(t) \quad \Rightarrow \quad X(s) = \frac{F(s) + (ms + b)x_0 + mv_0}{ms^2 + bs + k}$$

Denominator: system dynamics. Numerator: applied force and initial-state terms.

At rest, $x_0 = v_0 = 0$. Then $X(s) = \frac{F(s)}{ms^2+bs+k}$;

Laplace transform: integration

CORE

Position accumulates velocity and retains its initial value

INTEGRATION IS ACCUMULATION

$$\mathscr{L}\left\{\int_0^t v(\tau)\,d\tau\right\} = \frac{V(s)}{s}$$

If velocity $v(t)$ is known, position is $x(t) = x_0 + \int_0^t v(\tau)\,d\tau$. The extra x_0 matters: **the integral starts from zero, but the physical position need not**.

WORKED EXAMPLE

Let $v(t) = e^{-2t}$m/s and $x_0 = 0.3$m. First transform $v(t)$, then divide by s to accumulate the travel.

$$V(s) = \frac{1}{s+2}, \quad X(s) = \frac{0.3}{s} + \frac{1}{s(s+2)} = \frac{0.3}{s} + \frac{1}{2}\left(\frac{1}{s} - \frac{1}{s+2}\right)$$

Thus $x(t) = 0.3 + 0.5\left(1 - e^{-2t}\right)$. Check: $x(0) = 0.3$ m and $x(\infty) = 0.8$m; the added 0.5m is the area under $v(t)$.

PART 2

Inverse Laplace transform: algebra to response

01 Inversion workflow

02 Lecture 02A method examples

03 Tank response; initial and final values

Inverse Laplace transform:

FROM $F(s)$ BACK TO $f(t)$

$$F(s) = \mathscr{L}\{f(t)\} \quad \Longrightarrow \quad f(t) = \mathscr{L}^{-1}\{F(s)\}$$

1. Put $F(s)$ into a form that matches known transform pairs.
2. If necessary, split a rational expression into simpler terms.
3. Invert each term, then check the initial and final behaviour.

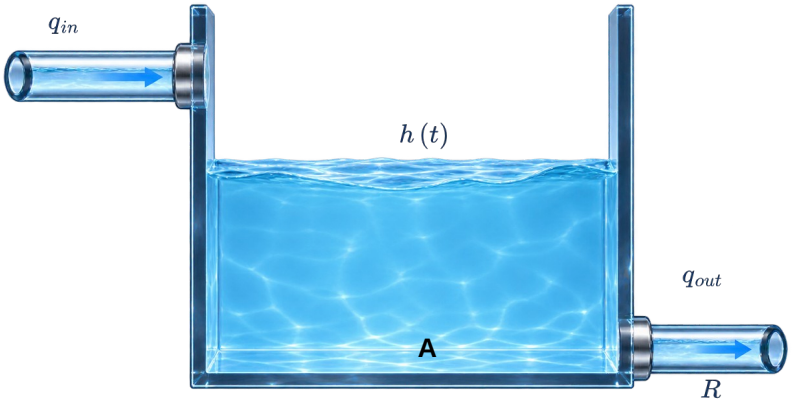
Required self-study: Lecture 02A

Distinct and repeated poles; quadratic factors. Long division: extension.

Tank problem: step inflow

QUESTION

THE PHYSICAL PROBLEM



A pump begins feeding an initially empty tank at $t = 0$. The sketch shows a later level $h\,(t)$. Water accumulates over area A; outlet flow rises with level: $q_{out} = \frac{h}{R}$.

Given $A = 2$ m², $R = 4$ s/m² and $q_{in}\,(t) = 0.1u\,(t)$ m³/s. Initially $h\,(0) = 0$.

MODEL FROM LECTURE 1

$$A\frac{dh}{dt} = q_{\text{in}}\,(t) - \frac{h\,(t)}{R} \quad \Rightarrow \quad 2\frac{dh}{dt} + \frac{h}{4} = 0.1u\,(t)$$

Find $h\,(t)$, its final level and time constant. Then ask what changes if the pump is off but the tank starts at a nonzero level h_0.

Tank solution: level response

ANSWER

TRANSFORM THE BALANCE WITH $h(0) = 0$

$$A\left[sH(s) - h_0\right] + \frac{H(s)}{R} = Q_{\text{in}}(s), \quad Q_{\text{in}}(s) = \frac{0.1}{s}$$

$$(2s + 0.25)H(s) = \frac{0.1}{s} \quad \Rightarrow \quad H(s) = \frac{0.1}{s(2s + 0.25)}$$

PARTIAL FRACTIONS AND INVERSION

$$H(s) = \frac{0.4}{s} - \frac{0.4}{s + 0.125}$$

$$h(t) = 0.4\left(1 - e^{-t/8}\right) \text{ m}$$

Checks: $h(0) = 0$; $h(\infty) = Rq_{\text{in}} = 0.4$ m; time constant $AR = 8$. At $t = 8$ s the level has reached about 63% of its final rise.

Pump off, $h(0) = h_0$: $H(s) = \frac{h_0}{s+0.125}$, so $h(t) = h_0 e^{-t/8}$. The **initial-value term** supplies the entire draining response.

Initial-value theorem

Starting output immediately after switching, f(0+).

$$f\left(0^{+}\right)=\lim_{s\to\infty} sF\left(s\right)$$

Multiply by s; take $s\to\infty$.

Requires a finite starting output and no output impulse.

Initial value: switching on the tank pump

EXAMPLE

QUESTION

Tank on slides 16–17: find the level immediately after switching on the pump.

$$H(s) = \frac{0.4}{s(1+8s)}$$

Multiply by s; high-s limit:

$$\lim_{s\to\infty} sH(s) = \lim_{s\to\infty} \frac{0.4}{1+8s} = 0$$

The stored water level is continuous at switching.

Final-value theorem

Constant output approached after the transient.

$$f(\infty) = \lim_{s \to 0} sF(s)$$

Multiply by s; take $s \to 0$.

Rational transform: all poles of $sF(s)$, after cancellation, must lie in the left half-plane.

Final value: the tank reaches a steady level

EXAMPLE

QUESTION

Tank on slides 16–17: find the settled level with the pump running.

$$H(s) = \frac{0.4}{s(1+8s)}$$

The factor s cancels the step term; $s \to 0$ gives final value.

$$\lim_{s \to 0} sH(s) = 0.4$$

At equilibrium, outlet flow equals inlet flow.

PART 3

Mechanical examples: model to response

01 Lecture 1 Example 1: established model and task

02 Insert data, transform and invert

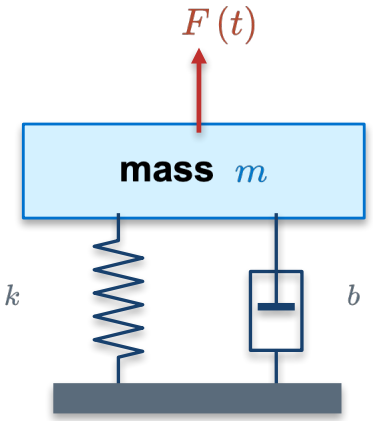
03 Free vibration and release-condition checks

Mass–Spring–Damper Response

CORE

STARTING MODEL FROM LECTURE 1 EXAMPLE 1

$$2\ddot{x} + 8\dot{x} + 80x = 2, \qquad x(0) = 0, \quad \dot{x}(0) = 0$$



fixed support

x upward from equilibrium

Fixed support; 2 N upward step; initially at rest at equilibrium. Use the Lecture 1 coordinate and parameters.

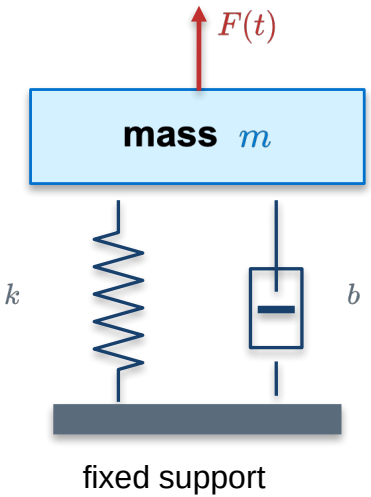
Task

Find the displacement response by Laplace transform and partial fractions; plot the result.

Lecture 1 Example 1: Laplace solution

EXAMPLE

$m = 2$ kg, $b = 8$ N·s/m, $k = 80$ N/m; *a 2* N *step acts from rest*



fixed support

1 Start from the established numerical ODE

$$2\ddot{x} + 8\dot{x} + 80x = 2, \qquad x(0) = 0, \quad \dot{x}(0) = 0$$

2 Transform and isolate $X(s)$

$$2\left[s^2X(s) - sx(0) - \dot{x}(0)\right] + 8\left[sX(s) - x(0)\right] + 80X(s) = \frac{2}{s}$$

Substitute $x(0) = 0$ & $\dot{x}(0) = 0$ into above equation

$$X(s) = \frac{1}{s\left(s^2 + 4s + 40\right)}$$

Damped spring–mass: invert $X(s)$

EXAMPLE

3 Decompose $X(s)$

$$X(s) = \frac{1}{s\left(s^2+4s+40\right)} = \frac{a_1}{s} + \frac{a_2 s + a_3}{s^2+4s+40} \qquad = \frac{(a_1+a_2)s^2 + (4a_1+a_3)s + 40a_1}{s\left(s^2+4s+40\right)}$$

$$\begin{cases} a_1 + a_2 = 0 \\ 4a_1 + a_3 = 0 \\ 40a_1 = 1 \end{cases} \qquad a_1 = \frac{1}{40}, \qquad a_2 = -\frac{1}{40}, \qquad a_3 = -\frac{1}{10}$$

4 Invert and check $x(t)$

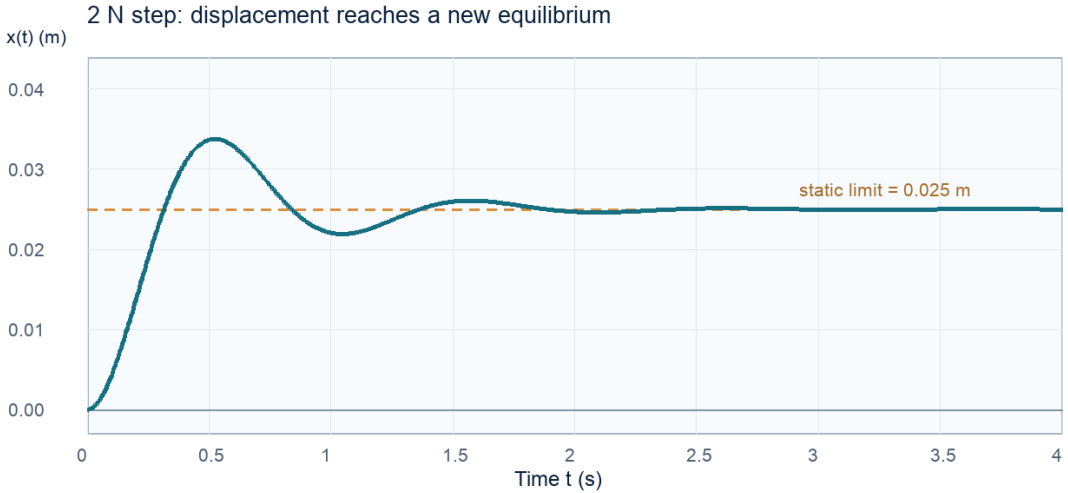
$$X(s) = \frac{1}{40s} - \frac{s+4}{40\left(s^2+4s+40\right)} = \frac{1}{40s} - \frac{s+2+2}{40\left[(s+2)^2+6^2\right]}$$

$$x(t) = \mathscr{L}^{-1}\{X(s)\} = \frac{1}{40} - \frac{e^{-2t}}{40}\cos 6t - \frac{e^{-2t}}{120}\sin 6t$$

Check: $x(0) = 0$, $\dot{x}(0) = 0.$; $x(\infty) = \frac{1}{40} = 0.025$ m. Static check F/k = $\frac{2}{80}$ m.

Damped spring–mass: plot and check

EXAMPLE



MATLAB: REPRODUCE IT

```
t = linspace(0,4,400);
x = 1/40 - exp(−2*t).* ...
  (cos(6*t)/40 + ...
   sin(6*t)/120);
plot(t,x,'LineWidth',1.6);
grid on; xlabel('t (s)');
ylabel('x (m)');
```

Natural time scale

$\omega_n = \sqrt{\frac{k}{m}}$; $T_n = \frac{2\pi}{\omega_n}$. Undamped references; damping alters the oscillation.

Read the graph

Starts at zero; settles at 0.025 m after **overshoot and decaying oscillation**.

Free vibration: nonzero initial displacement

EXAMPLE

Question Find the displacement after a mass is released from rest 0.10 m below static equilibrium.

Mass: 1 kg Damping: 4 N·s/m Stiffness: 40 N/m

Displacement is positive downward from static equilibrium, so gravity cancels. No external force acts after release.

1 Write the initial conditions

$$x(0) = 0.1, \quad \dot{x}(0) = 0, \quad \ddot{x} + 4\dot{x} + 40x = 0$$

2 Transform every derivative, including its initial-value term

$$\left[s^2 X(s) - 0.1s\right] + 4\left[sX(s) - 0.1\right] + 40X(s) = 0$$

$$\left(s^2 + 4s + 40\right) X(s) = 0.1s + 0.4$$

Free vibration: inverse transform and checks

EXAMPLE

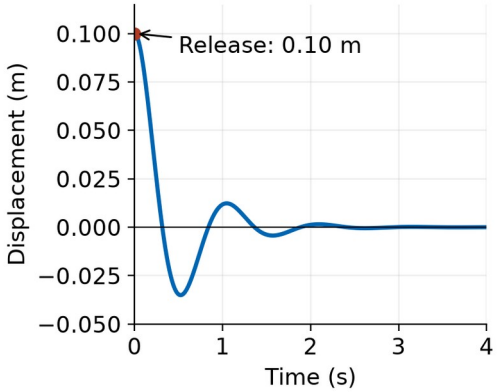
3 Match the shifted cosine and sine pairs

$$X(s) = \frac{0.1(s+4)}{(s+2)^2 + 6^2}$$

Separate the cosine and sine numerators:

$$0.1(s+4) = 0.1(s+2) + 0.2$$

$$X(s) = 0.1\frac{s+2}{(s+2)^2+6^2} + \frac{1}{30}\frac{6}{(s+2)^2+6^2}$$



4 Invert and check the release conditions

$$x(t) = 0.1e^{-2t}\cos(6t) + \frac{1}{30}e^{-2t}\sin(6t)$$

$$x(0) = 0.1, \quad \dot{x}(0) = -0.2 + \frac{6}{30} = 0$$

$$\lim_{t\to\infty} x(t) = 0, \quad s = -2 \pm 6j$$

Rotational extension: impulse torque

OPTIONAL

A fixed-axis rotor with viscous bearing drag

Problem

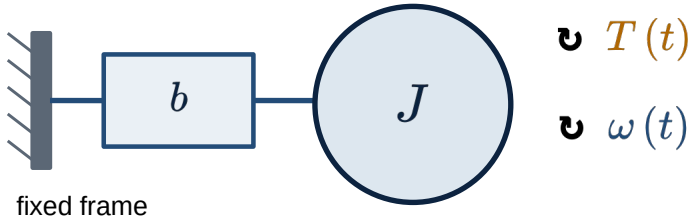
Rotor $J = 10$ kg·m²; bearing torque $-b\omega$, $b = 2$ N·m·s/rad. From rest, a clockwise 30 N·m·s torque impulse acts at $t = 0$. Find $\omega(0^+)$ and $\omega(t)$.

MODEL AND TRANSFORM

$$J\dot{\omega} + b\omega = T, \quad \omega(0-) = 0$$

or $10\dot{\omega} + 2\omega = 30\delta(t), \quad \omega(0-) = 0$

$$\Omega(s) = \frac{30}{10s + 2} = \frac{3}{s + 0.2}$$



INVERSE AND PHYSICAL CHECK

$$\omega(t) = \mathscr{L}^{-1}\left\{\frac{3}{s + 0.2}\right\} = 3e^{-0.2t}$$

Impulse jump: $\frac{30}{J} = 3$ rad/s. Decay time: $\frac{J}{b} = 5$ s.


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